

BlinkIT Grocery Sales Analysis Dashboard

1. Project Title - BlinkIT Grocery Sales Analysis Dashboard Using Power BI

Tools Used: Power BI, DAX, Microsoft Excel

Project Type: Data Analytics / Business Intelligence

Dataset: BlinkIT Grocery Data

2. Introduction

The BlinkIT Grocery Sales Dashboard is an interactive Business Intelligence project developed using Microsoft Power BI.

The objective of this project is to analyse grocery sales data and transform raw transactional data into meaningful business insights. The dashboard provides an overview of sales performance based on item categories, item fat content, outlet size, outlet location type, and outlet establishment year.

Interactive slicers allow users to filter the dashboard and analyze different segments of the business.

3. Project Objectives

The main objectives of this project are:

1. Analyze overall grocery sales performance.
 2. Identify the highest-performing item categories.
 3. Compare sales based on item fat content.
 4. Analyze sales across different outlet sizes.
 5. Compare sales across Tier 1, Tier 2, and Tier 3 locations.
 6. Analyze sales trends based on outlet establishment year.
 7. Create interactive KPIs using DAX measures.
 8. Build an easy-to-understand business dashboard using Power BI.
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4. Dataset Description

The project uses the **BlinkIT Grocery Data.xlsx** dataset.

The dataset contains:

- **8,523 records**
- **12 columns**
- **1,559 unique item identifiers**
- **10 unique outlets**
- **16 item categories**

Dataset Columns

Column	Description
Item Fat Content	Indicates whether the product is Low Fat or Regular
Item Identifier	Unique identifier of the product
Item Type	Category/type of grocery item

Column	Description
Outlet Establishment Year	Year in which the outlet was established
Outlet Identifier	Unique identifier of the outlet
Outlet Location Type	Location classification such as Tier 1, Tier 2, and Tier 3
Outlet Size	Size classification of the outlet
Outlet Type	Type of outlet/supermarket
Item Visibility	Percentage/value representing product visibility
Item Weight	Weight of the product
Sales	Sales generated by the product
Rating	Customer rating for the product

5. Data Preparation

Before creating the dashboard, the dataset was loaded into Power BI and prepared for analysis.

The preparation process included:

- Importing the Excel dataset into Power BI.
 - Checking the available columns and data types.
 - Reviewing missing values.
 - Verifying numerical fields such as Sales, Item Weight, Item Visibility, and Rating.
 - Checking categorical fields such as Item Type, Outlet Size, Outlet Location Type, and Item Fat Content.
 - Creating DAX measures for the dashboard KPIs.
 - Creating visualizations based on the prepared dataset.
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6. Power BI Dashboard

The dashboard was designed to provide both **high-level business KPIs** and **detailed sales analysis**.

The main sections of the dashboard are:

KPI Cards

The dashboard contains four KPI cards:

- Total Sales
- Total Items Sold
- Average Rating
- Average Sales

The current dashboard displays:

KPI	Dashboard Value
Total Sales	219.87K
Total Items Sold	2K
Average Rating	4.35
Average Sales	141.03

Note: These values should be verified against the DAX measures before the final report is submitted because the raw Excel dataset produces different overall figures.

7. DAX Measures

DAX measures were created in Power BI to calculate the key performance indicators dynamically.

The dashboard uses measures for:

Total Sales

Total Sales = SUM('BlinkIT Grocery Data'[Sales])

Average Sales

Average Sales = AVERAGE('BlinkIT Grocery Data'[Sales])

Average Rating

Average Rating = AVERAGE('BlinkIT Grocery Data'[Rating])

For **Total Items Sold**, the correct measure depends on what you intend "items sold" to represent because the dataset does not contain a quantity column.

For example, if you want the number of transactions/records:

Total Items = COUNTROWS('BlinkIT Grocery Data')

If you want the number of unique products:

Total Items = DISTINCTCOUNT('BlinkIT Grocery Data'[Item Identifier])

You should use whichever definition matches your dashboard objective.

8. Dashboard Visualizations

8.1 Item Type Analysis

A horizontal bar chart was created to compare sales across different grocery categories.

The major categories include:

- Fruits and Vegetables
- Snack Foods
- Household
- Frozen Foods
- Dairy
- Canned
- Baking Goods
- Health and Hygiene
- Meat
- Soft Drinks
- Breads
- Hard Drinks
- Starchy Foods
- Breakfast
- Others

From the raw dataset, **Fruits and Vegetables** and **Snack Foods** are the two largest sales categories.

8.2 Item Fat Content Analysis

A donut chart was used to analyze sales according to item fat content.

The two categories are:

- Low Fat

- Regular

A second visualization compares sales by fat content across different outlet location tiers.

This allows users to understand whether product fat content has any noticeable relationship with sales performance.

8.3 Outlet Size Analysis

A pie chart was created to compare sales across outlet sizes:

- Small
- Medium
- High

This visualization helps identify which outlet size contributes the largest share of sales.

8.4 Outlet Location Analysis

Sales were analyzed across three outlet location tiers:

- Tier 1
- Tier 2
- Tier 3

The dashboard currently shows:

Location Dashboard Sales

Tier 3 78.2K

Tier 1 72.4K

Tier 2 69.3K

According to the current dashboard, **Tier 3 outlets have the highest sales contribution.**

8.5 Sales by Outlet Establishment

A line chart was created to analyze sales according to the outlet establishment year.

This visualization helps identify:

- Sales growth
- Sales decline
- High-performing establishment years
- Changes in outlet performance over time

The chart makes it easier to identify years with unusually high or low sales.

9. Interactive Filters

The dashboard provides interactive slicers for:

Item Type

Users can select a specific grocery category and analyze its performance.

Outlet Size

Users can filter between:

- High
- Medium
- Small

Outlet Location Type

Users can filter between:

- Tier 1
- Tier 2
- Tier 3

These filters dynamically update the dashboard visuals and KPI measures.

10. Key Insights

Based on the dashboard:

Sales Performance

The dashboard provides a consolidated view of grocery sales performance through KPI cards.

Product Categories

Snack Foods and Fruits & Vegetables are among the strongest-performing categories in the dashboard.

Outlet Location

Tier 3 outlets currently show the highest sales contribution in the dashboard.

Outlet Size

Medium-sized outlets represent the largest sales contribution among the outlet-size categories shown.

Fat Content

The dashboard allows comparison between Low Fat and Regular products and their contribution to sales.

Outlet Establishment

Sales vary across establishment years, indicating differences in performance between older and newer outlets.

11. Dashboard Design

The dashboard was designed with a BlinkIT-inspired color scheme using:

- Yellow
- Green
- Orange
- Black/dark text

The design uses:

- Rounded containers
- KPI cards
- Consistent spacing
- Interactive slicers
- Bar charts
- Donut charts
- Pie charts
- Line charts
- Data labels
- Clear chart titles

The goal was to maintain a balance between **visual appeal** and **analytical readability**.

12. Business Questions Answered

The dashboard helps answer the following business questions:

1. What is the overall sales performance?
 2. Which grocery item categories generate the highest sales?
 3. Which outlet size contributes the most to sales?
 4. Which outlet location tier performs best?
 5. How do Low Fat and Regular products compare?
 6. How does sales performance vary across outlet establishment years?
 7. What is the average customer rating?
 8. What is the average sales value?
 9. How does performance change when different filters are applied?
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13. Technologies Used

Microsoft Power BI

Used for:

- Data visualization
- Dashboard development
- Interactive filtering
- Data analysis

DAX

Used for:

- KPI calculations
- Aggregations
- Dynamic measures

Microsoft Excel

Used as the source dataset.

Git & GitHub

Used for:

- Version control
 - Project documentation
 - Portfolio presentation
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14. Project Structure

BlinkIT-PowerBI-Dashboard/

```
|
|— README.md
|— BlinkIT Dashboard.pbix
|— BlinkIT Grocery Data.xlsx
|
|— screenshots/
|   |— dashboard.png
|   |— filters.png
|   |— kpis.png
```

└─ visual-1.png

└─ visual-2.png

└─ documentation/

└─ BlinkIT Power BI Project Report.pdf

15. Conclusion

The BlinkIT Grocery Sales Dashboard demonstrates how Power BI can be used to transform raw grocery sales data into an interactive analytical solution.

The project combines data preparation, DAX measures, KPI development, interactive filters, and multiple visualization techniques to provide insights into product categories, outlet characteristics, sales performance, and customer ratings.

The dashboard can help business users quickly identify high-performing product categories, outlet segments, and sales trends, supporting more informed business decisions.

16. Future Improvements

The dashboard could be further enhanced by adding:

- Year-over-year sales growth
- Sales contribution percentage
- Top 5 and Bottom 5 products
- Outlet-level performance
- Profit/margin analysis if cost data becomes available
- More advanced DAX measures
- Drill-through pages
- Tooltip pages
- Dynamic titles
- Sales forecasting
- Automated data refresh